The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III_1 Verdict

Harald Rønning

Abstract. A companion paper formulates the state problem at the center of this series: on the CCR structure produced by the two-contraction tower, find a state invariant under the surviving $SO(3)$ remnant and KMS with respect to the squeeze flow — the $\hbar$-preserving shadow of the octonionic grading dilation. We solve this problem in outline, with every load-bearing step proved, computationally verified, or cited and flagged. First, a no-go theorem: no such state — indeed no squeeze-invariant normal state at all — exists on any finite number of CCR fibers, and no invariant quasi-free covariance exists for the diagonal squeeze on any number of static fibers; the obstruction runs exactly through $W \neq 0$ and evaporates classically. Type-III necessity is thereby proven inside the model rather than imported. Second, the escape: the fiber squeeze is unitarily conjugate to translations along the log-radial line — the scale continuum itself becomes the mode space of second quantization, with spectrum absolutely continuous and equal to $\mathbb{R}$, forced rather than chosen. Third, the construction: a chiral, current-type quasi-free β-KMS state for translations over the scale line, rotation-invariant automatically, with $W = \hbar \cdot 1$ uniform. Fourth, the type: the closed group generated by the modular spectral data is computed directly — the forced interval spectrum saturates $\mathbb{R}$, while the counterfactual state coupled to the discrete grading ladder, run on the octonionic weights $(1, 2, 3)$, freezes on a lattice, giving type III_λ with $\lambda = e^{-\beta E_0}$ — the contraction scale imprinted. Together with the centralizer condition — established for this construction in a companion note — the factor of the squeeze-thermal state is the unique hyperfinite type III_1.

---

1. Introduction

The companion paper (*The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2*) ends by posing a precise problem. The two-contraction tower terminates in the $3{+}3{+}1$ Heisenberg algebra with $\{X^a, P_b\} = -2\delta^a_b W$ and W central; by the centrality theorem of this series and Schur's lemma, $W = \hbar \cdot 1$ in any irreducible representation, fixing the CCR fiber. Two structures are inherited from the octonionic level: the squeeze flow — the unique $\hbar$-preserving component of the grading dilation, with generator $D = \frac{1}{2}\sum_a (X_a P_a + P_a X_a)$ — and the $\mathfrak{so}(3)$ remnant of G_2, acting as one rotation on X and P simultaneously. The state problem: find ω that is KMS for the squeeze flow and invariant under the remnant. Three structural theorems shape the problem from outside (all cited in the companion paper): dynamics must be state-borne (Alfsen–Shultz), no algebra-resident density can generate the flow at a type III target (Takesaki), and the type of the resulting factor is a computation, not a premise.

This note solves the problem in outline. Statuses are labeled throughout: *proved* (here), *verified* (computationally, this series' standard), *standard* (textbook technology), *cited* (external result, flagged for verification), *open*.

2. The conjugacy: the squeeze is translation along the scale line

Proposition 1 (proved; verified). Let $V : L^2(\mathbb{R}_+, dr) \to L^2(\mathbb{R}, du)$ be $(Vf)(u) = e^{u/2} f(e^u), u = \log r$. Then V is unitary and intertwines the squeeze flow $(U_t f)(r) = e^{t/2} f(e^t r)$ with rigid translations $u \mapsto u + t$. On the full fiber the same mechanism holds with the three-dimensional Jacobian: $W : L^2(\mathbb{R}^3) \to L^2(\mathbb{R}_u) \otimes L^2(S^2)$, $(Wf)(u, \theta) = e^{3u/2} f(e^u \theta)$, is unitary for the measure $r^2\, dr\, d\Omega$ and carries the squeeze $(e^{itD} f)(x) = e^{3t/2} f(e^t x)$ to translation $u \mapsto u + t$ identically on every angular channel, with generator $-i\partial_u$. The translated line is the *radial scale* coordinate $u = \log r$, not a Cartesian axis of the Heisenberg fiber: what the flow shifts is the continuum of scales.

Verification. The intertwining is an exact change-of-variables identity, confirmed numerically to defect 1.2×10^{-15}; unitarity (the Jacobian half-power) confirmed by quadrature, $\|Vf\|^2 = 0.7106535$ against $\|f\|^2 = 0.7106546$, the difference being finite-window tail truncation.

Corollary 1.1. The spectrum of D is purely absolutely continuous and equal to $\mathbb{R}$ — *forced* by the conjugacy, not chosen.

Corollary 1.2 (verified). The squeeze commutes with the diagonal $SO(3)$ exactly: at the symplectic level both are block-scalar, and $\|[S_t, R \oplus R]\| = 0$ to machine precision. Rotations act purely on the angular factor.

3. The no-go: the state cannot exist in type I, and W is the reason

Proposition 2 (proved). On any finite number of fibers, no normal state is invariant under the squeeze flow; a fortiori, no KMS state exists there.

Proof. A normal invariant state is a positive trace-one operator ρ commuting with e^{itD}. Trace-one implies compact, so ρ's spectrum away from zero is pure point with finite-rank spectral projections; these commute with e^{itD}, yielding finite-dimensional invariant subspaces on which D has eigenvalues — contradicting Corollary 1.1 (the total generator over n fibers retains purely absolutely continuous spectrum). Hence $\rho = 0$, contradicting trace one. KMS implies invariance, so no KMS state. ■

Proposition 3 (proved; verified). For the diagonal squeeze acting on *any* number n of static fibers, no invariant quasi-free covariance exists. The space of invariant symmetric forms was solved exactly (dimension 1 for $n = 1, 9$ for $n = 3$, all cross-pairings included): *every* solution has identically vanishing x–x block. The quasi-free positivity bound $\eta(x,x)\,\eta(p,p) \geq \frac{1}{4}\sigma(x,p)^2$, with $\sigma(x,p) = -2W$, then reads $0 \geq W^2$: violated for every $W \neq 0$, and vacuous exactly at $W = 0$.

Remark (the mechanism). The no-go is strictly quantum. The structural $\hbar$ — whose survival through two contractions this series engineered — is itself what expels its own thermal flow from every type-I setting. This is the internal counterpart of Takesaki's criterion (semifinite $\iff$ modular flow inner): the sought state can exist only where density matrices fail. Proposition 3 also identifies the escape route: invariance fails for any flow that *squeezes static modes*; it can hold only for a flow that *translates modes along a continuum* — and Proposition 1 shows the fiber supplies exactly that continuum, the scale line itself.

4. The construction

Take the one-particle space $H_1 = L^2(\mathbb{R}_u) \otimes L^2(S^2)$ with one-particle generator $h = -i\partial_u$ (Proposition 1). Perform the chiral positive-frequency restriction — the $p > 0$ half of each translation line, the $p < 0$ half entering as conjugates [standard CCR technology for generators with two-sided spectrum]. Handle the $p \to 0$ infrared point by current-type smearing, test functions carrying a p-factor, as for the chiral $U(1)$ current [standard chiral bookkeeping; careful execution in these variables remains to be written out]. On the resulting Weyl algebra take the quasi-free β-KMS state ω_β for the translation flow [existence: standard for this class; the thermal-chiral-net literature is the anchor, cited pending verification]. Then:

- ω_β is KMS for the squeeze flow, *by Proposition 1* — the flow is the same operator in different coordinates;
- ω_β is $SO(3)$-invariant *automatically*: its covariance is a function of h, which commutes with rotations (Corollary 1.2);
- $W = \hbar \cdot 1$ uniformly: one central scale for all modes, as the centrality theorem licenses;
- nothing is inserted by hand: the algebra came from the tower, the flow from the dilation's $\hbar$-preserving shadow, the mode continuum from Proposition 1, and the state from the KMS condition itself.

Every constraint of the state problem is met.

5. The type computation

The invariant [standard, Connes]: $S(M) \setminus \{0\}$ is a closed multiplicative subgroup of $\mathbb{R}_+^\times$; the trichotomy $\{1\}$ / $\lambda^{\mathbb{Z}}$ / $\mathbb{R}_+^\times$ is the type trichotomy semifinite / III_λ / III_1.

The spectral data [standard; Araki: a quasi-free state is the unique KMS state of its Bogoliubov flow]: for the quasi-free KMS state at inverse temperature β, $\log \mathrm{Spec}(\Delta_\omega)$ is the closed additive group generated by $\beta \cdot \mathrm{supp}(h)$ under finite particle/hole insertions (signs $\pm$).

The computation (verified; grid resolution 10^{-5}, gaps measured on $[-3, 3]$, $\beta = 1$). The generated group was constructed directly for four spectral hypotheses:

spectral data	insertions	max gap	verdict signature
single energy 0.7 (Powers)	4 / 16 / 48	0.700 / 0.700 / 0.700	frozen lattice: III_λ
grading ladder $(1, 2, 3) \cdot 0.7$	6 / 24 / 48	0.700 / 0.700 / 0.700	frozen lattice: III_λ, $\lambda = e^{-0.7} \approx 0.4966$
two incommensurate $\{1, \sqrt{2}\}$	8 / 24 / 64 / 120	0.243 / 0.071 / 0.029 / 0.017	dense: III_1
interval $(0, 1]$ (**our case, forced**)	4 / 8 / 16 / 32 energies	0.345 / 0.048 / 10^{-5} / 10^{-5}	saturates $\mathbb{R}$: III_1

The interval row is the only one available to the squeeze-thermal state: Corollary 1.1 forces absolutely continuous spectrum on all of $\mathbb{R}$, so the generated closed group is $\mathbb{R}$ and $\mathrm{Spec}(\Delta_\omega) = [0, \infty)$.

The one cited link. $\mathrm{Spec}(\Delta_\omega) = S(M)$ requires the centralizer condition [Connes' criterion], standard for free thermal states [Araki–Woods 1963; Araki–Woods, *A classification of factors*; Bratteli–Robinson II]. This is the single link of the chain not re-derived here; it is established for this construction in the companion note *The Centralizer Is Trivial*, where the centralizer is proven trivial and factoriality follows as a corollary.

Verdict. With that link in place: the factor of the squeeze-thermal state is of type III_1 — and by Haagerup's uniqueness theorem, it is *the* hyperfinite III_1 factor. The destination is canonical; the octonionic origin lives entirely in the labeling: the flow that is thermal, the rotational symmetry the state respects, the scale line that became the mode space, and the $O(\mu)$ R-flux corrections as perturbative memory.

6. The fork, resolved into two theorems

The fork — type III_1 or type III_λ — does not dissolve; it splits into two precise statements, both computed. *Within* the squeeze-KMS formulation, III_1 is forced: the spectrum is an interval by Proposition 1, and intervals saturate. *Outside* it, the alternative is exactly characterized: a state coupled instead to the discrete grading ladder — the octonionic weights $(1, 2, 3)$ are commensurate — yields type III_λ with $\lambda = e^{-\beta E_0}$, a preferred scale in the modular spectrum, set by the contraction. The program's claim is therefore conditional in exactly one place, and that conditionality is its falsifiable content: **if thermal time runs along the inherited dilation flow, the algebra of the world is the unique hyperfinite type III_1 factor; if nature instead couples to the grading ladder, the modular spectrum carries a preferred octonionic scale.** Either outcome remembers the origin.

7. Scope

Three parts of the argument are carried by companion notes, stated precisely. (i) The centralizer condition of §5: the companion note *The Centralizer Is Trivial* proves the centralizer trivial, with factoriality derived as a corollary. (ii) Uniqueness: the companion note *The State Is Unique* establishes ω_β as the unique regular β-KMS state, with rotation-invariance derived rather than imposed. (iii) The functional-analytic setting — test space, Weyl algebra, one-particle structure, and the verified KMS analyticity — is given in the companion note *The Infrared Write-Out*. One naming caution for the record: Shlyakhtenko's *free* Araki–Woods factors — a distinct, free-probability construction — exhibit the same lattice-versus-line type trichotomy through their orthogonal representation's spectral data; this is cited as a parallel instance of the mechanism, not as evidence for the present construction.

Remark (no regulators). No cutoff, hard or soft, appears anywhere in this construction, and the verdict is itself the formal statement of that absence. The chiral restriction is a doubling, not a truncation — the $p < 0$ half enters as conjugates, and no modes are discarded. The infrared condition is a scale-free domain condition on test functions ($\hat{f}(0) = 0$), proven *necessary* in the companion note on the centralizer, not a mode removal: the spectrum remains $(0, \infty)$ untruncated, and no ultraviolet regulator is present or needed. The connection to the type is exact: a hard cutoff yields type I — excluded for this state by Propositions 2–3, through $W \neq 0$ itself — while a residual preferred scale yields type III_λ, the computed counterfactual of §§5–6; type III_1 is precisely the absence of any distinguished scale, including in time, since for III_1 the set of modular periods (Connes' T-invariant) is $\{0\}$. Consistently, the quantities well defined here without any regulator are the relative, modular ones — Araki's relative entropy is defined directly on type III algebras — while those that would require a cutoff (traces, density matrices, absolute entropies) are exactly the ones the no-go expels. The verdict is also robust rather than fragile in this respect: even a hypothetical hard infrared truncation of the spectrum to $[\epsilon, \infty)$ would leave the generated closed group equal to $\mathbb{R}$, since any interval of positive length generates it.

Finally, bibliographic details below are given only to the precision verified within this series' sessions; page numbers are included only where independently confirmed.

References

- Companion papers: *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G2*; *The Central Element of Algebraic R-Flux*; *The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict*; *The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty*; *The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line.*
- Connes, A., "Une classification des facteurs de type III," *Ann. Sci. École Norm. Sup.* 6 (1973).
- Araki, H., "Relative entropy of states of von Neumann algebras," *Publ. Res. Inst.*

Math. Sci. Kyoto Univ. 11 (1976).

- Araki, H. & Woods, E. J., "Representations of the canonical commutation relations describing a nonrelativistic infinite free Bose gas," *J. Math. Phys.* 4 (1963), 637–662.
- Araki, H. & Woods, E. J., "A classification of factors," *Publ. Res. Inst. Math. Sci. Kyoto Univ.* 4 (1968).
- Araki, H., "On quasifree states of CAR and Bogoliubov automorphisms," *Publ. Res. Inst. Math. Sci. Kyoto Univ.* 6 (1970), 385–442.
- Connes, A. & Størmer, E., "Homogeneity of the state space of factors of type III_1," *J. Funct. Anal.* 28 (1978), 187–196.
- Haagerup, U., "Connes' bicentralizer problem and uniqueness of the injective factor of type III_1," *Acta Math.* 158 (1987).
- Takesaki, M., *Theory of Operator Algebras II*, Springer (2003).
- Bratteli, O. & Robinson, D. W., *Operator Algebras and Quantum Statistical Mechanics II*, Springer (1981).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.